Analytica Chimica Acta 1415 (2026) 345682

Contents lists available at ScienceDirect

Analytica Chimica Acta

journal homepage: www.elsevier.com/locate/aca

Rethinking local spectral modelling: From per-query refitting to model libraries

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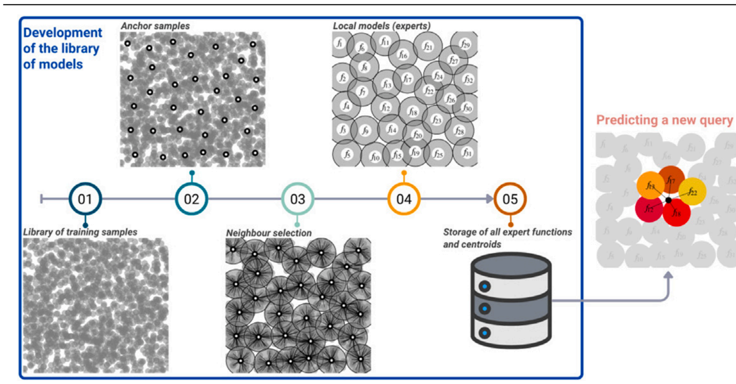
HIGHLIGHTS

- A new memory-based learning algorithmic framework, called `liblex`, is introduced.
- `liblex` constructs a library of local models rather than seeking a universal model.
- Predictions retrieve and ensemble pre-computed local models.
- `liblex` enables interpretable local models with per-sample uncertainty proxies.
- Performance is shown in a challenging cross-continental soil spectroscopy case study.

GRAPHICAL ABSTRACT



ARTICLE INFO

Handling Editor: Aoife Gowen

Dataset link: https://CRAN.R-project.org/package=resemble, https://github.com/l-ramirez-lopez/resemble, https://l-ramirez-lopez.github.io/resemble/

Keywords:

ABSTRACT

Background: This work introduces `liblex`, a simple yet powerful algorithmic framework to streamline the use of complex spectral libraries. It converts diffuse reflectance spectroscopy (DRS) libraries into libraries of pre-computed localised models (experts) that can be retrieved and combined at prediction time.
Results: The `liblex` algorithmic framework was evaluated in a challenging cross-continental test case. A large and heterogeneous DRS library of North American soils was used to construct a library of predictive models for total carbon (TC), which were then applied to samples from a region in the Democratic Republic of the Congo. In this scenario, `liblex` achieved high predictive accuracy and produced RMSE values that

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https://doi.org/10.1016/j.aca.2026.345682
Received 23 February 2026; Received in revised form 9 May 2026; Accepted 19 May 2026
Available online 25 May 2026

Local chemometrics
Diffuse reflectance spectroscopy
Retrieval-gated modelling
Cross-domain modelling
Transfer learning
Memory-based learning

were competitive with, and in this case lower than, those obtained with the benchmark methods considered, including LOCAL, Cubist, global PLS variants, and a convolutional neural network with transfer learning. This work also shows how `liblex` intrinsically provides uncertainty proxies via prediction dispersion and supports interpretability through stored regression coefficients and variable-importance profiles.

Significance: The `liblex` framework addresses several long-standing drawbacks of conventional local modelling methods by replacing per-query model refitting with the retrieval and aggregation of pre-computed localised experts. This design reduces computational burden at prediction time, enables deployment without access to the full reference library, preserves interpretability, and supports robust prediction across heterogeneous domains. The dispersion among retrieved expert predictions also provides an intrinsic, low-cost proxy for sample-specific uncertainty. These features make `liblex` a scalable, interpretable, and privacy-preserving strategy for operationalising large spectral libraries.

1. Introduction

The measurement and assessment of environmental, agricultural and food resources continues to rely heavily on conventional laboratory reference methods. Traditional analytical methods are highly accurate, however, they are typically slow, costly, and some are environmentally harmful. The fact that they rely on extensive sample preparation, specialised laboratory facilities, and chemical reagents creates a bottleneck in the efficient generation of data which is critically needed for decision-making and sustainable resource management. In this context, diffuse reflectance spectroscopy is an appealing alternative which can estimate multiple matrix properties simultaneously, in a rapid, non-destructive, and scalable fashion. This technique enables high-throughput data acquisition while reducing environmental impact and operational costs, meeting the growing demand for high-quality environmental and agricultural information [1,2].

Diffuse reflectance spectroscopy (DRS) does not replace conventional laboratory methods but is widely recognised as a complementary technique that enhances analytical throughput and scalability. Together, DRS and conventional methods enable the expansion of analytical data volume and resolution needed to meet the growing demand for quantitative information in environmental monitoring.

Although the return on investment in DRS technology is typically high compared to conventional methods (a fact repeatedly reported in the literature), it is also important to acknowledge that developing suitable training sets for a given application can be tedious, time-consuming, and expensive. Generally, a suitable training set should include a broad range of chemical and physical matrix conditions. However, factors such as long-term seasonal variations in compositional or physical characteristics, or even the seasonal availability of the target material, may substantially increase the time required to build an appropriate training set [3]. In addition, measuring the required response variables in the training set by conventional laboratory methods is also often cumbersome and expensive. Therefore, leveraging pre-existing training sets contained in spectral libraries provides an attractive alternative for the development of DRS for routine analysis.

Large, publicly accessible spectral libraries have become readily available in recent years, however, their potential is still far from fully realised. As spectral libraries expand, increasing heterogeneity translates into complex, context-dependent spectral–response relationships that make the use of simple global linear models challenging in practice. In soil DRS applications, for example [4], only a few research groups (typically those who built the libraries) have realised substantial benefit. Low adoption is largely due to the extensive domain adaptation required before models become operational in new contexts [5,6]. Importantly, the domain shifts encountered in practice may be substantial and context-dependent, rather than incremental, which is precisely what makes effective reuse of existing libraries challenging. The associated costs of adapting DRS library-derived models for operational use, including training set assembly and instrument acquisition, discourage adoption, particularly in resource-limited laboratories where analytical capacity is limited. Consequently, much of the value embedded in existing libraries remains inaccessible, slowing the integration of DRS into routine workflows. In this context, methods that exploit existing DRS libraries in a more effective way are crucial. They must reduce calibration costs and accelerate deployment (time-to-routine), so that spectral libraries can function as reusable operational assets rather than static repositories.

A common strategy to alleviate these challenges is *memory-based learning* (MBL; *a.k.a.* local modelling), which has a long history as an effective approach for managing large and heterogeneous spectral libraries [7]. In MBL methods, each query sample is paired with its most spectrally similar samples retrieved from a training set (library), together with their associated reference values. These locally selected samples are then used to fit a model specific to that query, which is subsequently used to predict its response value [8–11]. By restricting model fitting to locally relevant subsets of the library, MBL methods reduce the impact of global domain shifts and alleviate the need for extensive recalibration, making large spectral libraries more readily usable across diverse conditions. Numerous DRS studies report MBL as an accurate predictive method for large and heterogeneous libraries of different agricultural matrices [e.g. 12–18].

Established MBL approaches in DRS, such as locally weighted regression [19], LOCAL [8], and newer variants including kNN–Locally Weighted Partial Least Squares Regression [20] and locally weighted *roboost* regression [18], typically show strong predictive performance. However, MBL methods are characterised by several well-documented practical trade-offs. These include (*i*) potentially increased computational cost, as many MBL implementations rely on per-query similarity searches and on-the-fly fitting of local models, which (although algorithm- and implementation-dependent and often mitigable) may limit applicability in computationally constrained (e.g. embedded or microcomputer-based) or (near-)real-time settings; (*ii*) the lack of well-established, general-purpose interpretation frameworks for aggregating and comparing interpretability outputs across local, query-specific models; (*iii*) sensitivity to neighbour quality and distance metric choice, which, although often mitigated by local weighting, can, under heterogeneous or poorly covered domains, induce local extrapolation and increased uncertainty [21,22]; (*iv*) instability under spectral inconsistencies arising from sample processing differences or instrumental drift, which can distort neighbour selection and degrade predictive accuracy [7]; (*v*) the requirement to access the full reference library at prediction time, raising potential data-sharing and privacy concerns; and (*vi*) a conceptual limitation with respect to the memory-based learning analogy, in that similar spectra are retrieved, but the predictive solution is recomputed rather than explicitly stored and recalled. Spiers and Kalivas [23] provide an interesting discussion on some of these trade-offs. While methodological alternatives or custom remedies exist for many of these trade-offs, they are generally not intrinsic to the modelling framework and often require additional assumptions or implementation effort.

To address the trade-offs of MBL methods, this work proposes a reformulation of how both MBL and spectral libraries are operationalised. Rather than fitting ephemeral, query-specific local models from training spectra, the central idea is to construct a library of pre-computed local models that can be retrieved and combined for prediction, an approach referred to here as retrieval-gated ensembling. In this formulation, spectral libraries are transformed from passive repositories of training data into structured collections of reusable local experts/models, thereby eliminating the need to refit a model for each query. In this respect, the present work introduces a new algorithmic framework called `liblex`, which stands for *library of localised experts*. The framework exploits heterogeneous, large-scale spectral libraries by pre-calibrating local models and dynamically retrieving a small subset of relevant experts based on spectral similarity, whose predictions are combined through data-dependent weights. This design is related to mixture-of-experts frameworks, however, here the experts correspond to local linear models rather than neural network components. This model library design provides several practical advantages. Local experts are interpretable and computationally efficient once calibrated, enabling deployment without access to the full training dataset. The retrieval mechanism enhances robustness to spectral heterogeneity, while the dispersion among retrieved experts provides an explicit heuristic proxy for predictive uncertainty.

Conceptually, `liblex` can be viewed as a retrieval-gated form of a mixture-of-experts framework. In contrast to conventional mixture-of-experts models, where the gating function is typically learned parametrically, `liblex` selects and weights local experts using explicit spectral dissimilarities. Its connection to MBL lies in the use of local neighbourhoods in spectral space, but the predictive unit is an ensemble of pre-computed local linear experts rather than a single model fitted from neighbouring samples at prediction time. Model ensembling is built dynamically through spectral-similarity-based gating. In contrast to typical neural mixture-of-experts models, both the experts and the gating mechanism remain explicit and interpretable.

The central motivation of this work is therefore to formalise and evaluate `liblex` as a scalable framework for operationalising large, heterogeneous spectral libraries. This follows the principle articulated by Box [24], that states that all models are wrong but some are useful, and aligns with model averaging and multimodel inference, where predictions are formed by combining multiple candidate models to account for model uncertainty and improve predictive performance [25]. In this context, the specific aims of this research are to: (*i*) develop a framework that constructs libraries of pre-computed, localised experts from large spectral repositories; (*ii*) design a retrieval-gated mechanism that selects and integrates relevant experts according to spectral similarity, predictive quality, and uncertainty; (*iii*) present methods that produce interpretable model ensembles without requiring full-library access at prediction time; and (*iv*) benchmark the predictive performance and robustness of `liblex` relative to other state-of-the-art methods typically used in DRS.

2. Theory

The algorithmic framework introduced here addresses the challenge of exploiting large, heterogeneous spectral libraries to build accurate DRS models when target-domain reference data are scarce or unavailable. Let

$$\mathcal{D}_u := \{\mathbf{x}_u^i, y_u^i\}_{i=1}^m \tag{1}$$

denote the target set (where u indicates an "unknown" set), consisting of m paired observations $(\mathbf{x}_u^i, y_u^i)$ originating from the population of interest. Here, $\mathbf{x}_u^i \in \mathbb{R}^d$ represents the spectrum of the ith sample measured at d wavelengths or wavenumbers, and $y_u^i \in \mathbb{R}$ denotes the corresponding reference measurement of the response obtained by conventional laboratory analysis (e.g. dry combustion for total carbon). Note that y_u^i may be partially or entirely missing for the target population. Similarly, let

$$\mathcal{D}_r := \{(\mathbf{x}_r^j, y_r^j)\}_{j=1}^n \tag{2}$$

denote the spectral library (where the subscript r indicates that this library is a "reference" set), a pre-existing collection of n candidate samples for model training. This library is typically large and heterogeneous and does not necessarily contain observations from the target population, i.e. $\mathcal{D}_u \not\subset \mathcal{D}_r$. Furthermore, we assume that the target set $\mathcal{D}_u$ is representative of the target population and $n \gg m$.

2.1. The `liblex` algorithm: a retrieval-gated library of localised experts

The `liblex` algorithm is simple: it precomputes many small local linear models, referred to here as *experts*, and stores them in a library along with their centroids (i.e. the mean predictor vector). Prediction for a query sample (or query dataset) then reduces to retrieving (gating) a small number of experts with low centroid dissimilarity to the query sample, followed by a weighted average of their predictions, thereby avoiding per-sample refitting. This strategy reduces prediction variance while preserving locality. Fig. 1 illustrates a simplified, hypothetical example of `liblex`. In the left panel, samples are represented as points in a two-dimensional space. A set of 32 anchor samples (white circles) define 32 local experts, each obtained by fitting a local linear model around its anchor and jointly covering most of the whole space. For a new query sample (right panel), prediction proceeds by retrieving the nearest experts, where the position of each expert corresponds to its centroid. The individual predictions from these experts are then combined through a distance-based weighted average to produce the final output.

2.1.1. Construction of local experts

Building on the spectral library $\mathcal{D}_r$, `liblex` constructs a library of localised experts. Each expert is a linear local model calibrated on a neighbourhood comprising an anchor sample (i.e. a spectrum) and its $(k-1)$ nearest neighbours. In the present implementation, anchor samples are drawn from the training set. However, the framework only requires anchors to define neighbourhoods, so anchors could in principle be any spectra, including synthetic spectra. Let $\mathcal{A} \subseteq \{1, \ldots, n\}$ denote the anchor index set. For each $a \in \mathcal{A}$, define

$$\mathcal{N}_k(a) = \{a\} \cup \mathrm{NN}_{k-1}(a), \tag{3}$$

where $\mathrm{NN}_{k-1}(a)$ is the set of indices of the $k-1$ samples in $\mathcal{D}_r \setminus \{a\}$ most similar to $\mathbf{x}_r^a$ under a chosen metric. Note that the anchor a is explicitly included in the neighbourhood, so that $\mathrm{NN}_{k-1}(a)$ denotes the $k-1$ other nearest samples and $|\mathcal{N}_k(a)| = k$. The expert associated with anchor a is then fitted using only those samples in $\mathcal{N}_k(a)$ for which the response value is observed:

$$\{\, (\mathbf{x}_r^j, y_r^j) \,:\, j \in \mathcal{N}_k(a),\ y_r^j \text{ is observed} \,\}. \tag{4}$$

If y_r^a is observed, the anchor participates in the fit; otherwise it contributes only to neighbourhood definition. Consequently, `liblex` naturally accommodates missing responses in the anchor samples of the training spectral library.

Each anchor sample generates an expert or linear model, and each expert is stored in the library of models together with the predictor centroid, which is defined by

$$\mathbf{c}_a = \frac{1}{k} \sum_{j \in \mathcal{N}_k(a)} \mathbf{x}_r^j \tag{5}$$

Note that these are the empirical centres of the neighbourhood $\mathcal{N}_k(a)$, computed from all k members, and they do not necessarily coincide with the anchor spectrum $\mathbf{x}_r^a$, which only serves to define the neighbourhood. The neighbourhood centre captures the central tendency of all neighbourhood members, reducing sensitivity to a noisy or unrepresentative anchor. Because only neighbourhood centroids and models (rather than individual training spectra) need to be retained, this representation also facilitates privacy-preserving deployment by avoiding direct access to samples from the reference training library.

Ideally, anchor samples should be selected so as to provide adequate coverage of the spectral space represented by the training library, thereby avoiding poorly represented regions. The simplest way to guarantee full coverage is to designate every sample in the training library as an anchor, however, this substantially increases computational and storage costs. Conversely, selecting too few anchors may lead to an underrepresentation of the heterogeneity in the spectral–response relationships, particularly in sparsely populated regions of the library.

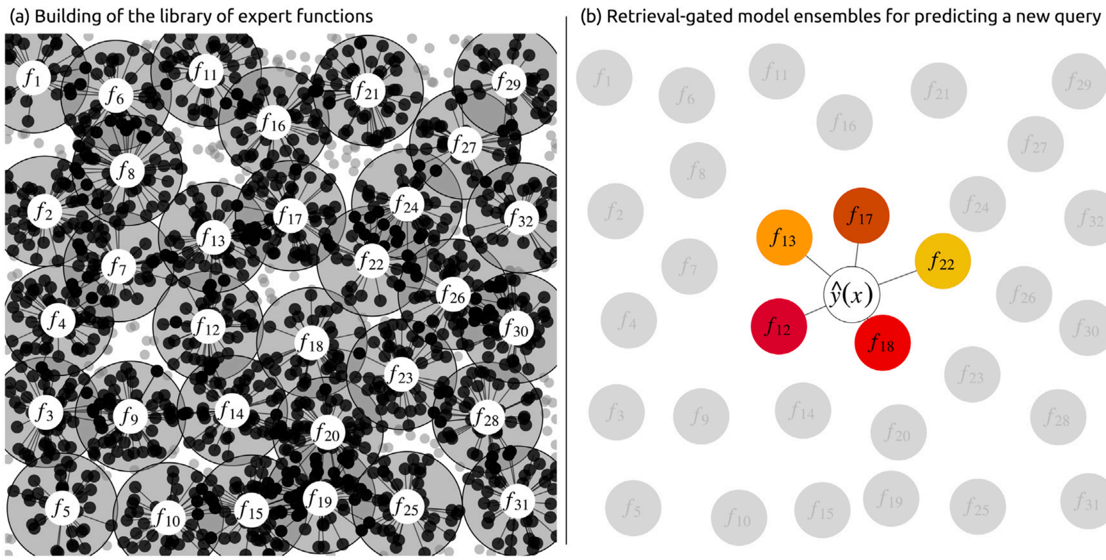

Fig. 1. Hypothetical schematic of the `liblex` workflow (32 anchors). (a): Each anchor (white circle) defines a fixed-size k-neighbourhood (grey circle); lines join neighbourhood members to their anchor. Each neighbourhood produces one expert/function so that the final library of experts is of size 32. (b): When a query sample is presented (hollow-outlined circle) its h nearest experts are retrieved, and the final prediction $\hat{y}(x)$ results from the combined predictions of its nearest experts, $\hat{y}(x) = \sum_{a \in \mathcal{H}(x)} w_a(x)\, f_a(x)$. Heat colours encode the importance of each expert for the prediction of the query (higher intensity indicates a greater gating weight $w_a(x)$ and hence a larger contribution to the final estimate $\hat{y}(x)$).

To have a good balance between coverage and efficiency, a structured sampling design can therefore be employed to select a representative subset of anchor samples.

Local regressions in `liblex` use a locally weighted average PLS (waPLS; 8). For each anchor a and neighbourhood $\mathcal{N}_k(a)$, waPLS constructs a model by aggregating PLS models fitted with different numbers of latent components $\kappa \in \{\kappa_{\min}, \dots, \kappa_{\max}\}$. Each κ produces a PLS model characterised by a vector of regression coefficients and a latent-space projection of the spectra, from which spectral reconstruction errors can be computed. Models coming from lower numbers of latent components are penalised because of their limited ability to accurately reconstruct the spectral signal, despite producing smooth and stable regression coefficients. Conversely, models with larger numbers of latent components typically provide better spectral reconstruction but are increasingly penalised, as higher-order PLS components lead to less smooth and less stable regression coefficients. This weighting strategy reduces sensitivity to the choice of the upper and lower bounds of κ. Algorithmic details of waPLS are provided in Appendix A. The weighting method employed here to aggregate different PLS components follows Shenk et al. [8], although other weighting strategies could also be employed.

2.1.2. Neighbour retrieval

Neighbour retrieval can use any predictor-space dissimilarity. In this case, a moving-window correlation dissimilarity was implemented. For two spectra $\mathbf{x}_r^j, \mathbf{x}_r^a \in \mathbb{R}^d$ and window length τ (e.g. 101 variables), let $n_\tau = d - \tau + 1$ and define

$$\operatorname{cor}(\mathbf{x}_r^j, \mathbf{x}_r^a; \tau) = \frac{1}{2n_\tau} \sum_{t=1}^{n_\tau} \Big[1 - \rho\big(\mathbf{x}^j_{r,t:(t+\tau-1)}, \mathbf{x}^a_{r,t:(t+\tau-1)}\big) \Big], \tag{6}$$

where $\rho(\cdot,\cdot)$ denotes the Pearson correlation coefficient. This reduces to $\frac{1}{2}\left[1 - \rho(\mathbf{x}_r^j, \mathbf{x}_r^a)\right]$ when $\tau = d$. The resulting dissimilarity ranges from 0 to 1, with values closer to 0 indicating more similar spectra. This pairwise, model-free metric is shape-oriented, making it invariant to additive offsets and positive scaling. Using a moving window preserves sensitivity to local spectral features, such as band shape and small peak shifts, which are often important in DRS. The $k-1$ smallest dissimilarities define $\mathrm{NN}_{k-1}(a)$ and hence the neighbourhood $\mathcal{N}_k(a)$. The moving-window correlation dissimilarity was therefore selected as a pragmatic default for this study, but this choice is not intrinsic to the `liblex` framework. Other predictor-space dissimilarities could be used for neighbourhood construction and expert retrieval.

2.1.3. Expert gating and ensembling for prediction

Gating directs each sample to be predicted (query) to a subset of relevant experts. Each expert is a local model $f_a : \mathbb{R}^d \to \mathbb{R}$, indexed by an anchor a and characterised by the centroid $\mathbf{c}_a$ of the neighbourhood it represents. For a given query sample, its dissimilarity to each $\mathbf{c}_a$ is computed to retrieve its h nearest experts. Each of these experts then generates an individual prediction, producing h predictions per query. Expert knowledge can also guide the retrieval of specific experts. For example, if it is known that an expert was trained on data from the same target domain as the query, that expert can be prioritised or always included in the retrieved expert set. The final prediction for the query sample is obtained by applying weights to the predictions of the selected experts and summing them.

Define $s_a(\mathbf{x}) := \operatorname{cor}(\mathbf{c}_a, \mathbf{x}; \tau)$ and let $\mathcal{H}(\mathbf{x})$ be the h experts with the smallest dissimilarities. Define the weights as

$$w_a(\mathbf{x}) = \frac{\exp\{-s_a(\mathbf{x})^2\}}{\sum_{b \in \mathcal{H}(\mathbf{x})} \exp\{-s_b(\mathbf{x})^2\}}, \tag{7}$$

and the final prediction is given by

$$\hat{y}(x) = \sum_{a \in \mathcal{H}(\mathbf{x})} w_a(\mathbf{x})\, f_a(\mathbf{x}) \tag{8}$$

Note that this requires only the stored centroids and model parameters, not the original training set.

In summary, `liblex` is governed by a small set of hyperparameters and design choices with interpretable effects on the behaviour of the framework. The neighbourhood size k controls the locality of each expert: smaller neighbourhoods produce more strongly localised experts but may be less stable and may leave gaps in the coverage of the predictor space, whereas larger neighbourhoods produce smoother experts but may introduce non-linear effects within neighbourhoods and increase overlap among local models. Some degree of overlap is desirable as it promotes smooth transitions between local models. However, excessive overlap may make neighbourhood centroids more similar to one another, which could reduce the discrimination of the retrieval step at prediction time. The anchor set $\mathcal{A}$ determines the

Table 1
Distribution statistics for total carbon (%) in the Congo target set and the North American soil spectral library.

Dataset	n	mean	median	sd	IQR	min	max	skew
Congo	737	1.26	1.16	0.58	0.71	0.37	4.74	1.30
North America	60272	5.24	1.46	10.85	2.98	0.05	59.97	3.14

number of localised experts, the coverage of the spectral space, and the storage and computational cost of the expert library. Too few anchors may underrepresent relevant regions of the predictor space, whereas too many anchors increase computational cost. Therefore, structured sampling designs may be useful for selecting well-distributed anchors. The number of retrieved experts h controls the degree of ensembling at prediction time, with smaller values favouring locality and larger values increasing smoothing and robustness. In this study, h was set equal to k, but the optimal relationship between neighbourhood size and the number of retrieved experts is likely data-dependent and warrants further methodological investigation. The window length τ controls the balance between sensitivity to local spectral structure and stability of the moving-window correlation dissimilarity used for neighbourhood construction and expert retrieval. Finally, the PLS component range $[\kappa_{\min}, \kappa_{\max}]$ controls the flexibility of each local regression. The locally weighted average PLS formulation used here reduces sensitivity to the exact number of retained components compared with standard PLS, but the selected range can still influence the stability and smoothness of the local experts. In this study, these parameters were either selected by validation or fixed pragmatically based on spectroscopic considerations; future work could usefully examine the sensitivity of these choices across datasets.

3. Methodology

The proposed method was evaluated using a large, heterogeneous infrared (IR) soil spectral library from the United States together with an independent target dataset from Tshuapa Province in the Democratic Republic of the Congo. The objective was to obtain simple, accurate models of total carbon (TC) for the target set. The IR measurements and routine laboratory analyses of TC were performed independently for the source and target datasets (two separate instruments of the same model, laboratories, acquisition dates, operators, and geographic domains).

This evaluation is challenging because a spectral library from one continent is used to predict an entirely independent set of samples from another continent, involving substantial differences in soil types, measurement protocols, instrumentation, and environmental conditions.

3.1. Datasets

3.1.1. Target domain: Tshuapa province, Democratic Republic of the Congo

The target population comprises 738 soil samples from the Tshuapa Province (Democratic Republic of the Congo) spanning forest and agricultural/rangeland sites. These samples were collected over an area of approximately 3305 km^2 and during three independent field campaigns [26–28]. All IR spectral measurements were conducted by Summerauer et al. [13] over the range from 7500 cm^{-1} to 600 cm^{-1}, at a resolution of 2 cm^{-1}.

Outliers were screened following De Maesschalck et al. [29]. Preprocessed spectra were projected onto the first five principal components (84% variance); samples with a Mahalanobis distance greater than 3 from the centroid were flagged, and one was removed.

The TC distribution for the target set (Table 1) is relatively narrow: mean 1.26% (median 1.16%), standard deviation 0.58%, interquartile range 0.71%, and observed range 0.37–4.74%. The skewness of 1.30 indicates a predominance of soils with low to moderate carbon contents and a limited number of higher-carbon samples.

3.1.2. Reference spectral library: North America

This publicly available dataset, compiled by the USDA-NRCS-NSSC Kellogg Soil Survey Laboratory (Lincoln, NE, USA) and available via the Open Soil Spectral Library (OSSL–KSSL v1; [30]), contains more than 68000 samples predominantly from the United States. Records include IR spectra, laboratory measurements, site metadata, and quality-control metrics. To harmonise sources, entries were retained only when spectrum–response IDs matched; duplicates were removed; incomplete spectra or anomalously high absorbances were excluded; TC values reported as zero were discarded (assumed below detection or unmeasured); and records outside the Americas were excluded. The resulting "North American IR spectral library" comprises 60272 samples (all with TC), largely from North America (only 78 from Central/South America: about 60 from Costa Rica, 10 from Suriname, and 1 from Chile), with spectra over 4000–600 cm^{-1} at 2 cm^{-1} resolution.

No outlier detection was applied to the North American library. Given its size and diversity, the library was assumed to provide adequate coverage of the target population. Occasional outlying samples were expected to have limited influence on the modelling results.

As expected, the North American library shows a much broader TC distribution than the Congo set, with a mean of 5.24% and a median of 1.46% (strong positive skew), spanning 0.05%–60% (Table 1). This heavy-tailed heterogeneity underscores the difficulty of directly transferring calibration models between these two domains.

3.2. Spectral data

Spectra were trimmed to 4000–605 cm^{-1} and resampled to a constant resolution of 8 cm^{-1}, returning a total of 425 spectral variables (Fig. 2). This resolution was chosen to retain explanatory power for TC while reducing dimensionality, as recommended by Deiss et al. [31] for the same type of data.

To minimise additive and multiplicative scatter effects, spectra were preprocessed with a Savitzky–Golay second-order derivative [32] followed by standard normal variate transformation [33]. The derivative used a window length of 25 spectral variables (corresponding to a span of 200 cm^{-1}) and a polynomial order of 2.

3.3. Model library construction and retrieval using `liblex`

Anchor samples: To reduce computational cost for the `liblex` experiments, experts were built only for an anchor subset while keeping all library samples eligible for neighbour retrieval. Anchors were chosen by comparing each Congo spectrum to the North American library using the moving-window correlation dissimilarity with $\tau = 101$. This choice of τ reflects a trade-off between local spectral sensitivity and the stability of the resulting moving-window correlation dissimilarity. For each Congo sample, the 200 most similar library spectra were retrieved, and the union of these candidate sets was deduplicated. This procedure produced 3467 unique anchor samples, which were then used to construct the library of experts. Accordingly, in this experiment the expert library is constructed in a target-oriented manner, tailored to the Congo prediction task; this design choice serves to reduce computation and is not intrinsic to the `liblex` framework, which can equally be applied in a target-agnostic setting.

From the Congo dataset, a small subset of $m = 20$ samples with available TC measurements was randomly selected and used as labelled target-domain references. These samples were pooled with the North American library and used (*i*) to participate in the calibration of local regressions and (*ii*) to act as anchors, thereby introducing a limited but explicit target-domain signal into the library. The choice of $m = 20$ reflects a realistic assumption that acquiring a small number of labelled target samples is feasible in practice and does not represent a major cost or time burden. The remaining 717 Congo samples, for which TC values were withheld, were used exclusively as unlabelled anchors to support neighbourhood definition and coverage of the Congo spectral space;

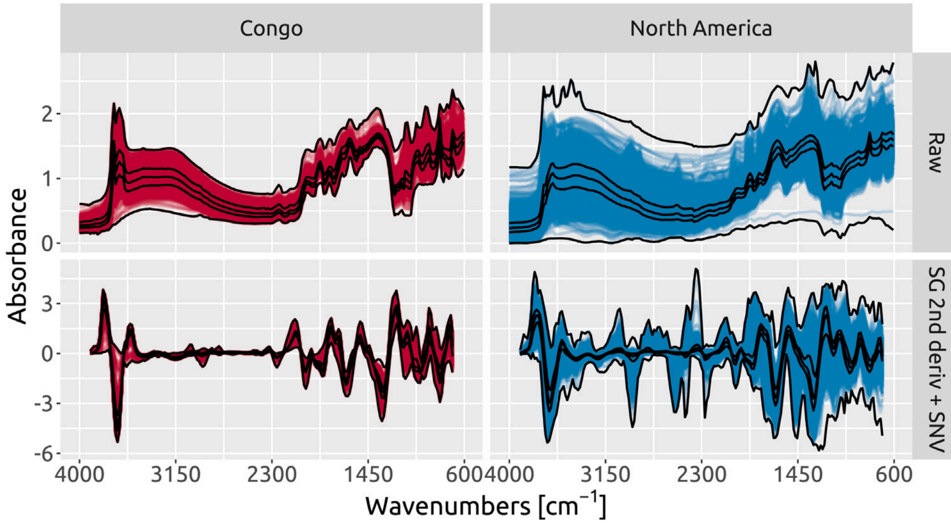

Fig. 2. Spectra of the Congo dataset and the North American soil spectral library. Black lines represent the 0%, 25%, 50%, 75%, and 100% quantiles.

they did not contribute to model fitting. This use of unlabelled spectra is consistent with semi-supervised spectroscopic modelling strategies, where unlabelled measurements are used to improve representation of sample or instrumental variability without requiring additional reference analyses [34]. The reference TC values of the 717 unlabelled Congo samples were kept fully independent and used only for final validation. In total, 4204 $(3467 + 20 + 717)$ anchor samples were used, combining labelled and unlabelled spectra to balance target-domain orientation with unbiased performance assessment.

Hyperparameter optimisation: The `liblex` algorithm was applied to the North American spectral library $\mathcal{D}_r$ pooled with the 20 labelled Congo samples to construct 4204 experts from the neighbourhoods of their anchor samples, which included the 717 unlabelled Congo samples as structural anchors. Sixteen different values of the neighbourhood size k were tested, ranging from 50 to 500 in increments of 30, with a separate expert library built for each k. Validation followed the nearest-neighbour cross-validation approach [9], in which, for each anchor with an observed response, a model is fitted excluding that anchor, and then that model is used to predict it. For waPLS, an exhaustive grid search was conducted over all possible component ranges with $2 \leq \kappa_{\min} \leq \kappa_{\max} \leq 30$, which resulted in 406 evaluated configurations. For each combination of k, $\kappa_{\min}$ and $\kappa_{\max}$, the RMSE across anchors with observed TC responses was computed, and the configuration that minimised RMSE was selected.

Expert gating: Expert retrieval was performed independently for each query sample. To guarantee that each prediction remained anchored to the target domain, all experts constructed from the Congo subset were always included in the retrieved neighbourhood. These target-domain experts were then given the same weight as the nearest centroid-matched expert from the library, ensuring they contributed comparably to the final prediction.

Validation and final evaluation: The hyperparameters k and $[\kappa_{\min}, \kappa_{\max}]$ were selected using nearest-neighbour cross-validation on the anchor-based expert library constructed from the North American library pooled with the 20 labelled Congo samples. Using this optimal configuration, a final library of localised experts was built. This fixed expert library was then applied to the remaining 717 Congo samples, which were not used in any stage of model fitting or hyperparameter selection. Predictions were obtained through sample-specific expert retrieval and gating, without refitting local models. Predictive performance was assessed by computing the RMSE against the reference TC values. In addition, the retrieved experts were analysed for variability in neighbourhood composition, regression coefficients, and variable importance profiles, the latter estimated using the selectivity ratio method [35].

3.4. Benchmark models and comparisons

To benchmark `liblex`, five additional modelling methods were used for comparison. The purpose of the benchmarking was not to attempt to claim overall algorithmic superiority (performance may depend on the dataset of course) but rather to provide context for interpreting the results. The aim was to show whether the proposed method performs on par with established approaches and does not fall noticeably behind them, thereby demonstrating that it offers competitive and practically useful performance.

PLS: The simplest benchmark was a global PLS regression developed exclusively with the 20 labelled samples from the Congo dataset. These were the same samples extracted from the Congo dataset and pooled with the North American library (see 3.3). This model represents the limiting case in which only a very small, domain-specific calibration set is available and no exploitation of existing libraries is attempted. Due to the very small sample size, the number of PLS components was optimised using leave-one-out cross-validation, with a maximum of 15 components. This upper limit was chosen to extend beyond the expected optimum, providing a check that cross-validation could detect overfitting at excessive model complexity.

PLS [Congo + North America]: This benchmark corresponds to a global PLS regression developed from a pre-filtered version of the North American library. A nearest-neighbour pre-filtering step based on the moving-window correlation dissimilarity was applied between each Congo sample and all samples in the North American dataset. Reference samples with a correlation dissimilarity above 0.5 were discarded, reducing the library to 13078 samples. These were then pooled with the 20 labelled Congo samples to develop a model for predicting TC. This benchmark represents the case where a large external library is used directly, without any sample selection beyond the initial pre-filtering step. The number of PLS components was optimised by withholding the 20 labelled Congo samples during fitting and using them only for component selection. For each candidate number of latent components, a PLS model was fitted on the 13078 North American samples and then applied to the labelled Congo subset. The optimal number of components was chosen as the one that minimised the RMSE on this target-domain subset. This strategy was adopted so that component selection was driven by target-domain predictive performance rather than by overall performance in the pooled data. After selecting the optimal number of components, the final PLS model was refitted using the 13078 North American samples together with the 20 labelled Congo samples.

LOCAL [adjusted]: The LOCAL algorithm [8] is a memory-based learning method that has been widely and successfully applied in DRS spectroscopy [36]. LOCAL fits local PLS models for each query sample based on its nearest neighbours in the reference library. In this study, two modifications were introduced to improve target-domain alignment: (i) the correlation dissimilarity used in LOCAL was replaced by the moving-window correlation dissimilarity with $\tau = 101$; (ii) the 20 labelled samples extracted from the Congo dataset were forced to be included in each of the neighbourhoods of the 717 unlabelled Congo samples, ensuring that each local model was at least partially informed by target-domain data. The number of neighbours k and the range of PLS components were optimised, with k varying from 50 to 500 in increments of 30 and the number of PLS components ranging from 2 to 30. In LOCAL, predictions are obtained from local regression models fitted using the same waPLS algorithm [8] as in `liblex`. A nearest-neighbour cross-validation approach [9] was used for hyperparameter optimisation.

Cubist: Cubist [37] is a rule-based regression tree algorithm that extends model trees with linear regressions at their terminal nodes. In addition to this global rule-based structure, Cubist employs a memory-based learning (MBL) component during prediction, where the model retrieves the k-nearest neighbours of a query case from the training set and uses them to adjust the rule-based regression estimate. This method has been successfully applied in various DRS applications, gaining notable popularity [22,38]. A grid search was conducted to optimise Cubist model hyperparameters, specifically the number of committees and neighbours. Committee sizes were varied from 1 to 100 in 40 evenly spaced steps, and neighbour values ranged from 0 to 9, resulting in 400 committee–neighbour combinations. For each configuration, a Cubist model was trained on the North American library after prefiltering with the moving-window correlation dissimilarity. The same 13078 prefiltered samples used in the PLS [Congo + North America] method were employed here. Model complexity was capped at a maximum of 1000 rules. Predictions were then generated for the 20 labelled samples from the Congo dataset. The optimal hyperparameter combination was identified as the one minimising the RMSE of predictions on these 20 labelled samples. The final model was then retrained using the optimal hyperparameters on the entire prefiltered North American library combined with the 20 labelled Congo samples. For final evaluation, the optimally configured Cubist model was applied to the 717 unlabelled samples from the Congo dataset.

CNN [North America, Congo adjusted]: A one-dimensional convolutional neural network (CNN) was trained on the North American library and later fine-tuned with 20 labelled Congo samples, representing transfer learning from a global to a target domain. The architecture comprised three convolutional blocks (Conv1D–BatchNorm–ReLU–MaxPool with 32, 64, and 128 channels, kernel sizes 5, 5, and 3, respectively) followed by two fully connected layers. This configuration follows the standard 1D CNN design widely used in spectroscopy, similar to Ng et al. [39], who employed stacked Conv1D–ReLU–pooling blocks for soil spectra, and to Passos and Mishra [40], who tested comparable multi-layer 1D CNNs for fruit quality prediction. Training employed AdamW optimisation with Smooth L1 loss, where β was set to 0.5 of the interquartile range of TC. Hyperparameters were selected via a basic grid search over learning rates $10^{-3}, 3\times10^{-4}, 10^{-4}$, weight decays $10^{-3}, 5\times10^{-4}, 10^{-4}$, and batch sizes $128, 256$. The restricted search was sufficient to ensure stable convergence. It was intentionally limited due to computational constraints and because the model was subsequently adapted using only a very small labelled target set, which reduces the value of extensive optimisation on the source domain. During adaptation, the convolutional backbone was frozen and only the linear layers were fine-tuned on the Congo samples, with early stopping based on validation loss. Final performance was assessed on the 717 held-out Congo samples.

3.5. External validation for modelling without target-domain adaptation

An additional evaluation was performed on the same external validation set under a restrictive setting in which no labelled Congo samples were available for model recalibration or adaptation. This scenario corresponds to model transfer to a new domain without access to any target-domain reference measurements. Although intentionally restrictive and less common in practice (where at least a small number of target-domain samples is typically available), it is included here to assess robustness under extreme domain shift.

In contrast to the experiments with limited target-domain adjustment, no Congo samples were used at any stage of model calibration, neighbourhood construction, expert selection, rule adjustment, or fine-tuning. This means that `liblex` operated without access to any target-domain labels, relying exclusively on its retrieval and aggregation mechanisms to generate predictions from the external library. All models relied exclusively on the North American spectral library, and method-specific mechanisms (such as memory-based adjustment in Cubist, local neighbourhood construction in LOCAL) operated only on the training data of the North American library. Hyperparameters and model structures were re-optimised under this constraint to ensure a fair comparison across methods. Note that in the case of the CNN, the same convolutional backbone and regression head previously trained only on the North American library was reused. For the PLS model and for Cubist, hyperparameters were optimised using 10-fold cross-validation. For LOCAL, hyperparameter optimisation was again performed using the nearest-neighbour cross-validation approach of Ramirez-Lopez et al. [9].

Once `liblex` and all benchmark models were optimised, they were applied to the 717 Congo validation samples to assess predictive performance under this no-domain-adaptation scenario.

3.6. Evaluation metrics

The performance of all the models was assessed using predictions for the 717 Congo samples. The following predictive accuracy metrics were used: coefficient of determination (R^2), root mean squared error (RMSE), and mean error (ME).

The benchmarking of `liblex` against the other methods was intended not to demonstrate general superiority over alternative models, but to provide context for interpreting its performance in the present case study. Nevertheless, to complement the descriptive comparison of RMSE values, pairwise comparisons of prediction errors were performed using van der Voet's randomisation test [41]. Briefly, for each benchmark method, its squared prediction error for each of the 717 validation samples was subtracted from the corresponding squared prediction error obtained with `liblex`, producing a vector of paired error differences. Under the null hypothesis of equivalent predictive accuracy, the signs of these paired differences are exchangeable. An empirical null distribution was therefore generated by randomly changing the signs of the paired differences over 9999 randomisations and recalculating the mean difference after each randomisation. The resulting p-value was computed as the proportion of randomised absolute mean differences that were at least as large in magnitude as the observed absolute mean difference, with the observed statistic itself included in both numerator and denominator following van der Voet [41]. Small p-values provide evidence against the null hypothesis of equivalent predictive accuracy, indicating that the observed reduction in RMSE for `liblex` is unlikely under the randomisation distribution of the paired squared prediction-error differences.

3.7. Implementation

All algorithms and experiments, except for the convolutional neural network, were implemented in `R` (v.4.5.1; R Core Team 42). The CNN was implemented in `Python` (v.3.11.9) using the `PyTorch` framework (v.2.5.1+cu121; 43). To address the computational demands of the algorithms implemented in R, core routines involving intensive matrix multiplications and iterative loops were implemented in C++ using the Armadillo library through RcppArmadillo, which extends the R–C++ interface provided by Rcpp [44,45]. Spectral preprocessing was carried out with the prospectr package [46]. The LOCAL algorithm for benchmarking was implemented using the resemble package [47].

The datasets, spectral preprocessing, and benchmarking framework employed here are aligned with those used in a companion study [48], which evaluates an alternative strategy for identifying optimal calibration subsets in spectral libraries for domain-adaptive calibration. This alignment was adopted to enable controlled comparison of distinct methodological approaches under identical experimental conditions.

4. Results and discussion

4.1. Performance with limited target-domain reference data

4.1.1. Model library construction and retrieval using liblex

Hyperparameters: The best hyperparameter configuration found corresponds to $k = 290$, $\kappa_{min} = 2$ and $\kappa_{max} = 30$. The nearest neighbour validation returned a $R^2 = 0.956$ and RMSE=0.353%.

Consistency: Fig. 3 shows that the centroids of the 4204 liblex models trained on the North American library closely resemble the Congo spectra. The main peaks and spectral patterns are similar, and the centroid envelope contains most Congo spectra, suggesting that the model library captures the dominant spectral variability of the target samples. Fig. 3 also shows that the centroids of the neighbourhoods used to train the expert models often differ from the anchor samples that initiated those neighbourhoods. This indicates that the local support retrieved around each anchor can be substantially displaced within the spectral space.

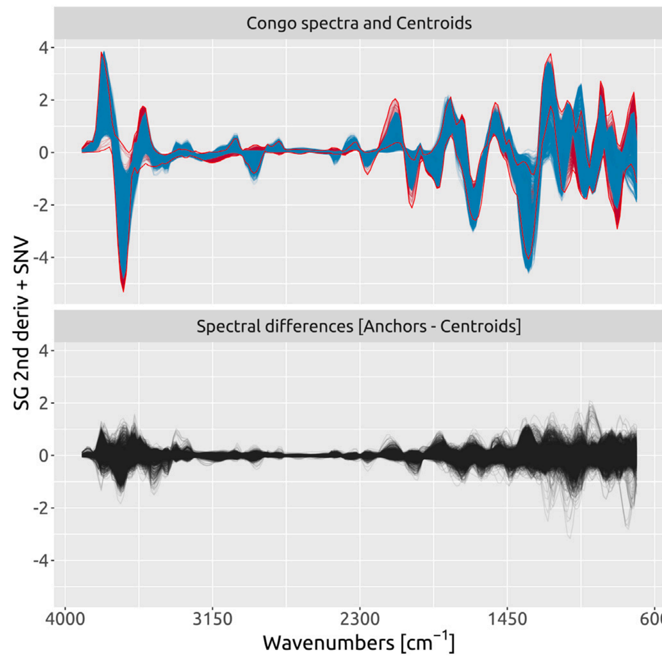

Fig. 3. Top: Centroids of the 4204 models developed with liblex (blue), with the spectra of the Congo samples shown in the background (red). Bottom: Spectral differences between each anchor sample and the corresponding centroid of the expert model it generated. On the y-axis, SG 2nd deriv. + SNV denotes Savitzky-Golay second-order derivative followed by standard normal variate processing. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Although centroid overlays suggest limited variation beyond the Congo spectra (Fig. 3), the 4204 liblex models exhibit marked heterogeneity in variable importance and regression coefficients (Fig. 4). In Fig. 4, grey spectra denote selectivity ratios for all the models, whereas red spectra highlight the selectivity ratios of the 100 models most frequently retrieved for Congo predictions. While the overall grey spectra show diverse, weakly aligned peaks, the red subset displays sharp, repeatable peak positions. The spectra of the regression coefficients are relatively smooth between 3150 cm^{-1} and 1704 cm^{-1}, compared to their left and right spectral edges (Fig. 4). These inconsistent patterns at the edges are probably due to the fact that the regression coefficients come from a weighted average of local models fitted with between 2 and 30 components. Although the waPLS algorithm penalises and down-weights noisy coefficients, models with a high number of components can still introduce some noise into the regression coefficients. In this context, averaging across models helps to mitigate this effect by reducing prediction variance and stabilising the resulting coefficient profiles. The shadowed spectra in the regression coefficient plots of Fig. 4 correspond to experts that fall outside the top 100 most frequently selected. Some of these experts may not have been selected at all. These infrequently used experts exhibit substantial variability across wavelengths, which is why their coefficient spectra are shown trimmed on the y-axis in Fig. 4. This variability in the regression-coefficient spectra and variable importance reflects the heterogeneity of the local models trained across the spectral library. Each expert is fitted on a different neighbourhood, and therefore its regression vector expresses a distinct local relationship between spectra and the target variable. The wide spread of these spectra suggests that the expert library may capture a broad range of local spectrum-response relationships arising from the diverse neighbourhoods represented in the spectral space, including relationships not directly observed in the Congo dataset alone. This broad coverage suggests that the method can accommodate new, unseen samples from the target domain. However, not all experts are equally relevant: some encode relationships that differ substantially from those found in the Congo data and are therefore rarely selected. In practice, the predictions are driven mainly by a subset of experts (those consistently retrieved).

Prediction performance and benchmarking: The liblex algorithm achieved a high predictive performance (see Table 2 and Fig. 5 for target-domain adaptation) with $R^2 = 0.842$, RMSE=0.243% and ME $= -0.008$%. In general, it produced competitive results compared to all its benchmarks (Table 2). Except for the PLS [Congo + North America] model, the remaining models showed comparatively low biases, indicating that target-domain adaptation was successful. The results of van der Voet's randomisation tests indicated that the observed RMSE differences between liblex and the benchmark methods are unlikely to be due to random variation in the paired prediction errors. For all benchmark methods except the CNN, van der Voet's randomisation tests provided strong evidence of differences in paired squared prediction errors relative to liblex ($p < 0.001$). For the CNN, the evidence was weaker but still significant at the 5% level ($p = 0.027$; Table 2).

Among the benchmark methods, the CNN achieved the second-best performance in terms of RMSE (Table 2 and Fig. 5 for target-domain adaptation). The optimal CNN configuration corresponded to a moderate model capacity, with 24 base channels in the initial convolutional layer, and used the Smooth L1 loss with the transition parameter set to $\beta = 0.5 \times$ interquartile range of the reference TC distribution.

For the adapted LOCAL model, the optimal configuration used 170 nearest neighbours, with the minimum and maximum numbers of PLS components set to 2 and 30, respectively. In terms of the RMSE, the adapted LOCAL model ranked third in performance (Table 2 and Fig. 5 for target-domain adaptation).

The Cubist model, using an optimal configuration of 67 committees and 9 neighbours, exhibited moderate predictive performance (Table 2 and Fig. 5 for target-domain adaptation).

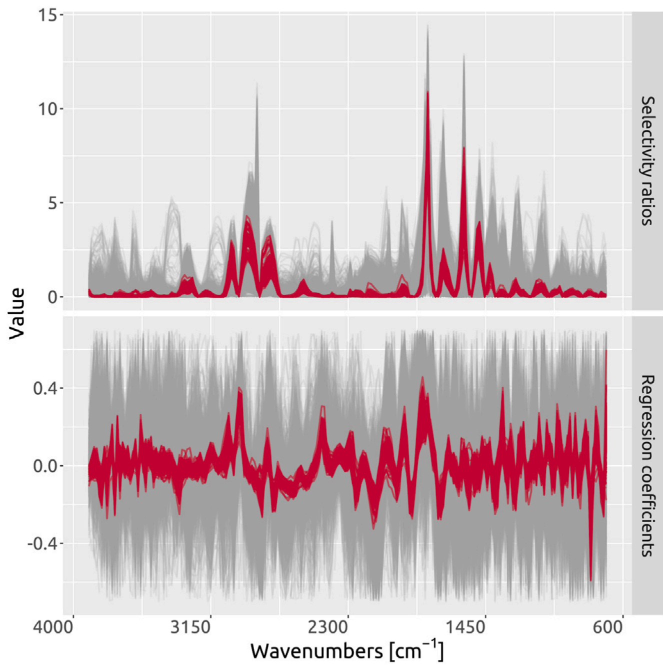

Fig. 4. Variable importance (*top*, selectivity ratios) and regression coefficients (*bottom*). Red spectra refer to the 100 most frequently retrieved models for Congo samples, while grey spectra show all 4204 models.

The PLS models calibrated using only the 20 Congo samples and those combining the Congo and North American libraries produced the poorest predictive performance (Table 2 and Fig. 5 for target-domain adaptation). The PLS [Congo] model achieved optimal performance with 8 components but remained limited by the small calibration set, resulting in comparatively high RMSE and low R^2. For the PLS [Congo + North America] model, the optimal number of components was 5 and it exhibited the highest error with pronounced bias indicating inadequate alignment between the source and target domains. However, it still achieved a relatively good R^2, indicating that, despite poor absolute accuracy, the predictions captured a substantial proportion of the variance in the target data.

The accuracy obtained with `liblex` is higher than that reported by Summerauer et al. [13] for the same region and dataset, where an RMSE of 0.319% was obtained using an African spectral library that included samples from neighbouring regions, implying less severe extrapolation than in the highly constrained setting analysed here.

It should be noted that replicate measurements or dataset-specific uncertainty estimates for the high-temperature dry combustion TC reference values were not available for the datasets used here. The reported RMSE values should therefore be interpreted as apparent prediction errors relative to the available laboratory reference values, rather than as pure model errors. In comparable soil applications, Stevens et al. [49] reported reproducibility errors of approximately 0.08–0.29% C for mineral soils analysed by dry combustion, while Even et al. [50] showed that soil C measurements can be strongly affected by sample-processing choices and inter-laboratory differences. The best RMSE values observed here are close to this reported reproducibility range, suggesting that part of the apparent prediction error may reflect uncertainty in the reference values themselves. Therefore, small RMSE differences between models, even when statistically supported by paired randomisation tests, should not necessarily be interpreted as evidence of practically meaningful superiority. Rather, the results indicate that `liblex` achieved competitive predictive performance relative to the state-of-the-art methods considered in this study.

The Congo validation set also has a relatively low mean TC value and a narrower TC distribution than the North American library, which affects the interpretation of both RMSE and R^2. The best RMSE values obtained here, approximately 0.24–0.31% TC, correspond to about 19%–25% of the mean TC value in the Congo set. However, the relatively modest R^2 values should not be attributed only to this restricted TC range; they likely reflect the combined effects of target-range restriction, reference-measurement uncertainty, complexity of the soil matrix, and the severe domain shift involved in transferring models from a broad North American library to a geographically and environmentally distinct Congo target domain.

Uncertainty proxy: An interesting feature of `liblex` is that predictions are weighted ensembles of retrieved expert outputs, which naturally provide a per-sample dispersion that can be used to approximate model uncertainty. This constitutes an advantage over the benchmark methods considered here, as none of them natively provide per-sample uncertainty proxies. For each individual prediction, its dispersion is the weighted variance of the predictions coming from the different experts used. Fig. 6 (left) shows that the residuals remain concentrated across most of the domain, but above values around 2%, the absolute residuals seem to increase at higher predicted TC values. Fig. 6 (middle) also shows that at TC values above 2%, there is a pronounced increase in the weighted standard deviation as predicted TC increases. This trend indicates that high-carbon samples occupy regions of the spectral space where the library provides less dense or more heterogeneous local support, resulting in greater disagreement among experts. It also suggests increasing spectral–response complexity at high carbon contents, where stronger absorptivity and non-linearities make local predictions more unstable. When comparing the absolute residuals and the weighted standard deviation of expert predictions in Fig. 6 (right), a positive relationship is observed, where predictions with low deviations tend to correspond to small errors, while large errors occur predominantly at high deviations. Together, these patterns suggest that expert disagreement is a coherent proxy for epistemic uncertainty (uncertainty arising from limited or heterogeneous coverage of the spectral space). In this case, for example, filtering observations with prediction standard deviation > 0.8 removed a single outlier, reduced RMSE to 0.229%, and increased R^2 to 0.857. Stricter thresholds reduce RMSE further but at the expense of coverage (See Appendix C). Importantly, this uncertainty metric is obtained intrinsically as part of the standard `liblex` workflow. This constitutes a practical advantage over many conventional memory-based or local modelling approaches, where uncertainty quantification is not available by default and is often introduced via ad-hoc resampling schemes or repeated neighbourhood refitting, which lead to substantial computational costs. While uncertainty estimation in local linear models is not inherently restricted to resampling (generic Bayesian formulations, approximate Bayesian inference, or analytical variance expressions can in principle be employed) these alternatives typically require additional modelling assumptions or customised implementations. In contrast, `liblex` natively provides per-sample proxies for predictive uncertainty without auxiliary inference procedures. However, this dispersion should be interpreted as a heuristic proxy rather than as a formal predictive interval or a calibrated estimate of total prediction uncertainty. Expert dispersion just reflects disagreement among the retrieved experts. So, it does not directly account for uncertainty in the laboratory reference measurements, systematic bias shared by all retrieved experts, or uncertainty introduced by spectral preprocessing and instrumental effects. It may also underestimate uncertainty when the retrieved experts are similarly miscalibrated. Therefore, expert dispersion should be interpreted as a diagnostic measure of local model disagreement rather than as a complete uncertainty quantification framework.

Emergent geographical pattern: The most frequently used expert models, defined as models used more than 10 times for any prediction in the Congo dataset, revealed an interesting geographical pattern. These models can be linked to their anchor samples in the North American library, which have associated geo-locations where these were collected. The vast majority of these anchor samples of frequently

Table 2
Predictive performance of `liblex` and benchmark models on the 717 validation samples under two experimental settings: *with target-domain adaptation* (20 labelled Congo samples available) and *without target-domain adaptation* (no Congo reference samples available). Benchmarks include LOCAL (adapted correlation dissimilarity; neighbourhoods forced to include target-domain samples when available), Cubist, PLS [Congo] (20 Congo samples), PLS [Congo + North America] (20 Congo samples plus 13078 North American samples), and a convolutional neural network (CNN; trained on North American samples and fine-tuned with 20 Congo samples when adaptation is permitted).

Model	R^2	RMSE (%)	ME (%)	p-value vs. `liblex`
With target-domain adaptation				
`liblex`	0.842	0.243	−0.008	–
Benchmarks:				
CNN [North America, Congo adjusted]	0.774	0.278	0.008	0.027[a]
LOCAL (adapted)	0.833	0.315	0.094	<0.001
Cubist	0.776	0.374	−0.001	<0.001
PLS [Congo]	0.670	0.403	0.062	<0.001
PLS [Congo + North America]	0.760	0.677	0.392	<0.001
Without target-domain adaptation				
`liblex`	0.794	0.277	−0.073	–
Benchmarks:				
CNN	0.469	0.628	0.295	<0.001
LOCAL	0.799	0.423	0.091	<0.001
Cubist	0.794	0.408	0.131	<0.001
PLS	0.814	0.978	0.737	<0.001

p-values are from van der Voet's two-sided randomisation test [41] comparing squared prediction errors of each benchmark against `liblex` (9999 permutations; minimum achievable $p = 0.0001$).

[a] Significant at $\alpha = 0.05$ but not at $\alpha = 0.01$.

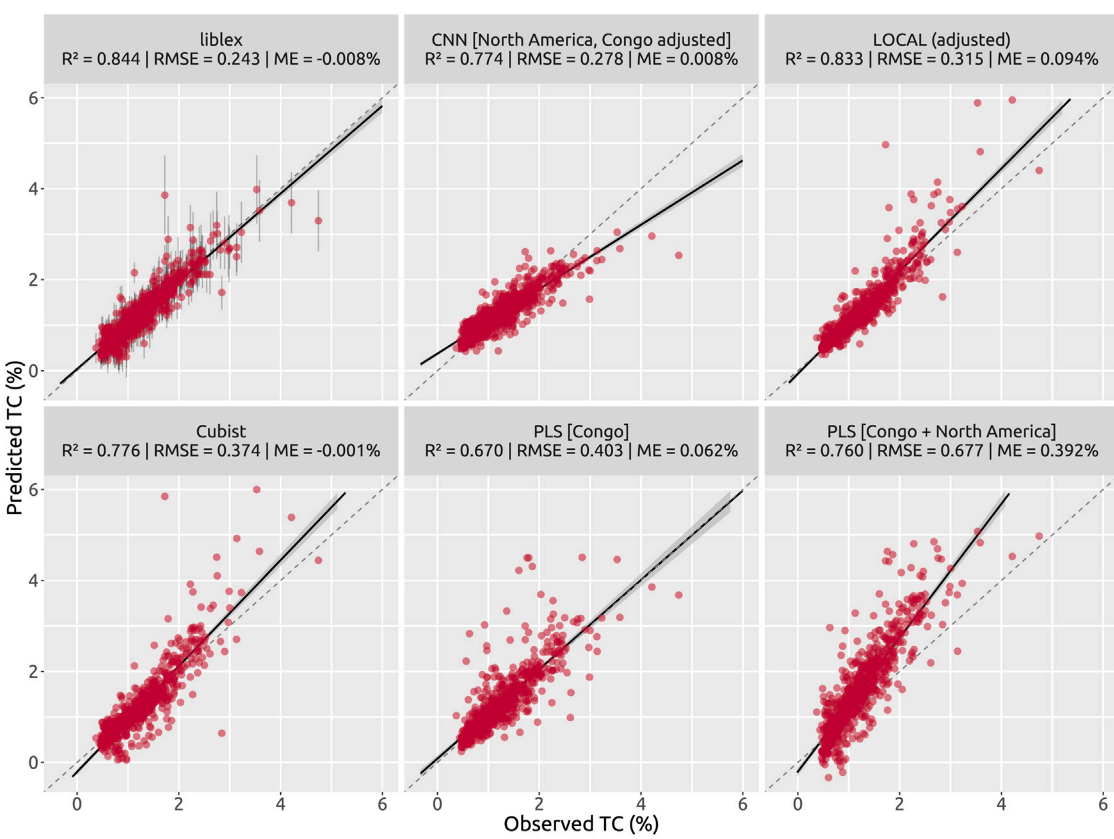

Fig. 5. Validation plots for `liblex` and the benchmark methods under the target-domain adaptation scenario. For `liblex`, the vertical bands indicate the weighted standard deviation across retrieved experts for each prediction.

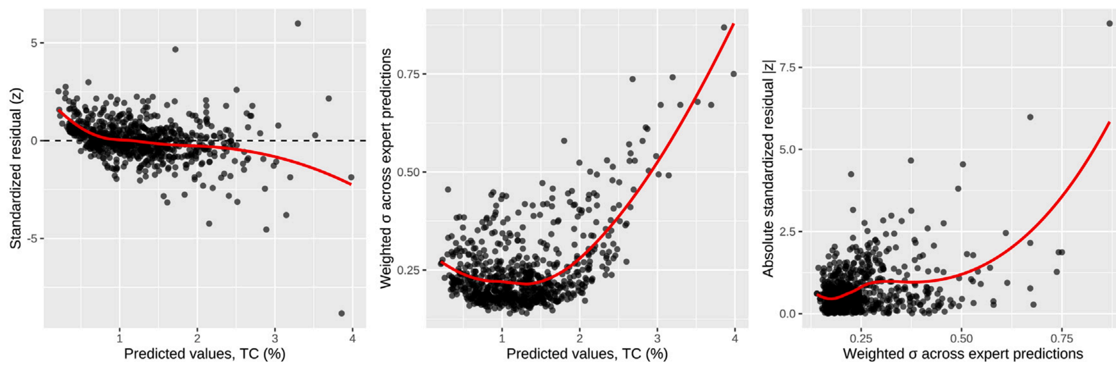

Fig. 6. Relationships between expert dispersion and prediction residuals in `liblex`, where residuals are defined as $e = TC - TC_{\text{predicted}}$.

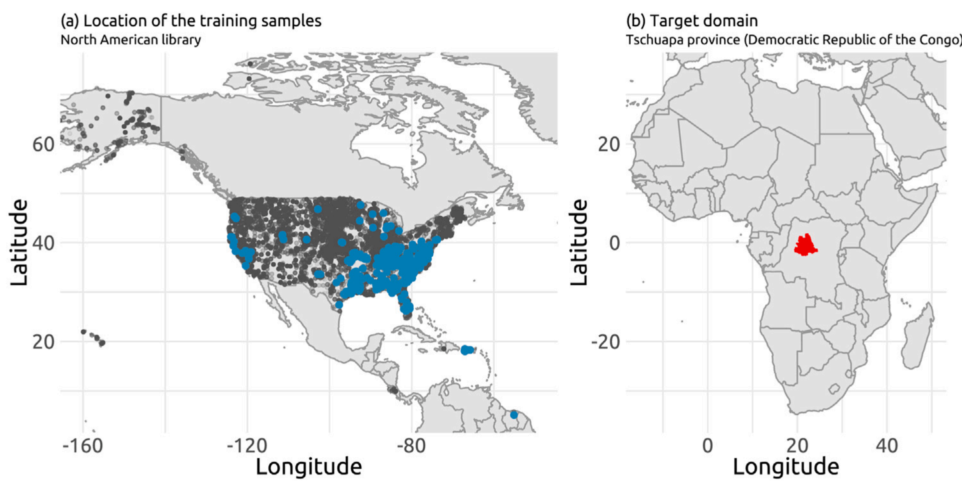

Fig. 7. (a): Geographic distribution of all the samples in the North American library (grey dots) and anchor samples that are associated to the experts used at least 10 times for the predictions (blue dots). (b): Tschuapa province (highlighted in red) in the Democratic Republic of the Congo, where the samples in the Congo dataset originate from. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

used experts originate from the southeastern United States (particularly the region spanning eastern Texas through the Lower Mississippi, the Gulf Coast, and up through the Carolinas) as well as from Puerto Rico (see Fig. 7, left). A plausible, though still speculative, explanation is that these frequently selected experts originate from regions with environmental and compositional characteristics that are partially aligned with those of the Congo samples. These observations indicate that `liblex` is able to retrieve experts that capture relevant domain structure, even when such relationships are not explicitly encoded in the model or provided as prior information.

4.2. Model performance without target-domain adaptation

In this evaluation, model performance was assessed in the absence of target-domain adaptation, with all models trained exclusively on the North American library and applied directly to the Congo dataset without access to any reference target-domain samples. As expected, the performance of all modelling approaches deteriorated in the absence of target-domain adaptation (Table 2, without target-domain adaptation), highlighting the importance of domain alignment. In this scenario, `liblex` produced the lowest RMSE and showed the least degradation in accuracy compared with the domain-adaptation case (Table 2, without target-domain adaptation). Pairwise comparisons using van der Voet's two-sided randomisation test indicated significant differences in paired squared prediction errors between `liblex` and all benchmark methods ($p < 0.001$; Table 2, without target-domain adaptation). Validation scatterplots are provided in Appendix B.

Except for the PLS model, for which the optimal number of latent components was 20, all other algorithms happen to arrive at the same optimal hyperparameters as in the target-domain adaptation scenario.

Beyond point accuracy, the absence of target-domain adaptation mainly affected calibration bias. The highest errors were observed for the CNN and PLS models. Their RMSEs appear to be driven by their large MEs (Table 2, without target-domain adaptation). Interestingly, the CNN model showed good predictive performance under the domain-adaptation scenario but performed poorly in the absence of adaptation, indicating a strong dependence on target-domain information in challenging domain transfer tasks. For PLS, RMSE increased slightly in the absence of target-domain samples compared with the model trained on the combined North American and Congo datasets (Table 2). In contrast, R^2 values remained similar across both settings, suggesting that both models captured comparable target-domain variability and that the effect of adding 20 Congo samples may have been diluted by the much larger North American training set. This indicates that the model preserved relative variation in the target set but did not achieve accurate calibration, whereas `liblex` achieved better bias control with only a modest reduction in variance explained.

The LOCAL and Cubist models showed similar performance under the no-adaptation setting (Table 2, without target-domain adaptation). For LOCAL, the absence of forced inclusion of target-domain samples in the neighbourhoods led to higher RMSE values, without a corresponding increase in ME. This suggests that the strategy of incorporating a small number of target-domain samples into each neighbourhood is important for ensuring a good prediction accuracy with LOCAL in a challenging domain transfer scenario. For Cubist, neighbourhood alignment did not substantially improve accuracy. When target-domain samples were unavailable and alignment relied solely on the North American library, the RMSE increased from 0.374% to 0.408% (Table 2).

4.3. Final considerations

The benchmarking results show that `liblex` performs competitively with established methods across a challenging cross-domain setting. This new method extends the toolkit available for local regressions and for modelling heterogeneous spectral datasets. By constructing a library of local expert models anchored in different regions of the spectral space, the method provides a structured way to explore local relationships, with interpretability supported by variable-importance profiles, regression-coefficient inspection and uncertainty proxies.

The approach introduced with `liblex` is also well suited to embedded or privacy-constrained settings. In many practical DRS applications, data protection requirements restrict the exchange of training datasets across institutions, as these may contain proprietary or sensitive information. One practical alternative is to exchange calibration models rather than the underlying training data. In this respect, `liblex` provides a framework through which data owners can do this. The design of `liblex` is consistent with current interest in federated and privacy-preserving learning [e.g.51,52], as it enables collaborative spectroscopic modelling while supporting institutional data sovereignty. In contrast to federated learning, which typically requires repeated communication rounds to train a shared global model, the `liblex` framework can operate through one-way sharing of locally trained models and on-demand retrieval at prediction time. Concretely, data owners can publish calibrated "expert" models, together with minimal metadata. This reduces the need for synchronised training schedules and iterative parameter exchange, without requiring access to the original training datasets.

An additional and potentially important implication of the `liblex` framework concerns its capacity for incremental extension. By design, the library of experts is modular, allowing new local experts to be added as new reference data or domains become available, without requiring systematic retraining of existing models. While this work focuses on a fixed expert library, such an incremental update strategy could substantially reduce recalibration costs, enable continuous anchoring to new conditions, and accelerate deployment in operational settings. Explicitly formalising update policies, compatibility assumptions between experts, and strategies for managing expert redundancy or obsolescence is an important direction for future research.

The `liblex` framework is conceptually related to the notion of receptive fields in locally weighted learning [53,54], where simple models are assumed to be valid only within local regions of the predictor space and are activated through distance-based gating. Like receptive-field methods, `liblex` combines the predictions of nearby local models to preserve locality. However, unlike classical receptive-field approaches that fit or adapt local models at prediction time, `liblex` constructs and stores a fixed library of local experts offline and performs prediction by retrieval and weighted ensembling. In addition, locality in `liblex` is defined through explicit neighbourhood construction and centroid-based expert representation, decoupling neighbourhood definition from model centring and enabling robust handling of heterogeneous libraries, missing responses, and structured expert reuse.

Another approach conceptually related to `liblex` is cluster-based modelling, in which the training data are partitioned into clusters according to predictor-space similarity and a separate predictive model is fitted for each cluster. In its standard formulation, each cluster is represented by a centroid or prototype, and prediction for a new observation is obtained from the model associated with the single closest cluster, yielding a hard assignment. Soft variants exist, in which predictions from multiple clusters are combined through distance-based weighting; however, clustering itself still relies on a prior hard partition of the predictor space into non-overlapping regions. In contrast, `liblex` does not impose a global partition of the space: neighbourhoods overlap by construction, and expert relevance is determined dynamically at prediction time through retrieval-gated weighting, resulting in a local and data-adaptive soft mixture-of-experts formulation.

5. Conclusions

This study introduced `liblex`, a retrieval-gated framework that operationalises large and heterogeneous spectral libraries by converting them into libraries of localised models, referred to here as experts. This method was used in a demanding cross-continental scenario, where a spectral library of North American soils was used to predict total carbon in soils of the Democratic Republic of the Congo. The use of `liblex` achieved high predictive accuracy and produced RMSE values that were competitive with those of state-of-the-art methods, including LOCAL, Cubist, global PLS variants, and convolutional neural networks. These results indicate that pre-computed expert libraries offer a viable alternative to conventional local modelling methods. In contrast to conventional memory-based or local modelling approaches, `liblex` does not require refitting a model for each query sample. Instead, predictions are obtained by aggregating pre-computed local experts, resulting in little computational cost of predictions.

By constructing a large collection of localised models in advance, `liblex` reformulates memory-based learning into a scalable and interpretable form of local chemometrics. The framework builds a retrieval-gated library of local PLS experts and generates predictions by ensembling the most relevant experts per query. Because experts are precomputed, prediction is computationally efficient, and each expert remains fully interpretable through its stored coefficients, variable-importance profile, and neighbourhood composition. Across the library, the experts exhibit diverse regression structures that reflect the heterogeneity of the underlying spectral space. Retrieval patterns reveal how different regions of the spectral space contribute to target-domain predictions, while the dispersion of expert outputs provides an intrinsic, low-cost proxy for uncertainty. These features provide a transparent view of how local relationships shape predictions and extend the interpretability of local chemometrics.

Overall, `liblex` provides a practical and conceptually simple mechanism for leveraging complex spectral libraries across domains. Future work should focus on optimising anchor selection and neighbourhood coverage, integrating adaptive update strategies as new data become available, and extending the framework to non-linear or hybrid expert architectures. Practical refinements include constraining PLS complexity, pruning redundant experts, and exploring the potential of other dissimilarity measures (e.g., metric learning or embedding-based distances) to improve neighbourhood construction beyond the moving-window correlation used here. By shifting the focus from retrieving samples to retrieving reusable experts, `liblex` provides a practical basis for scalable, interpretable, and collaborative local chemometric modelling.

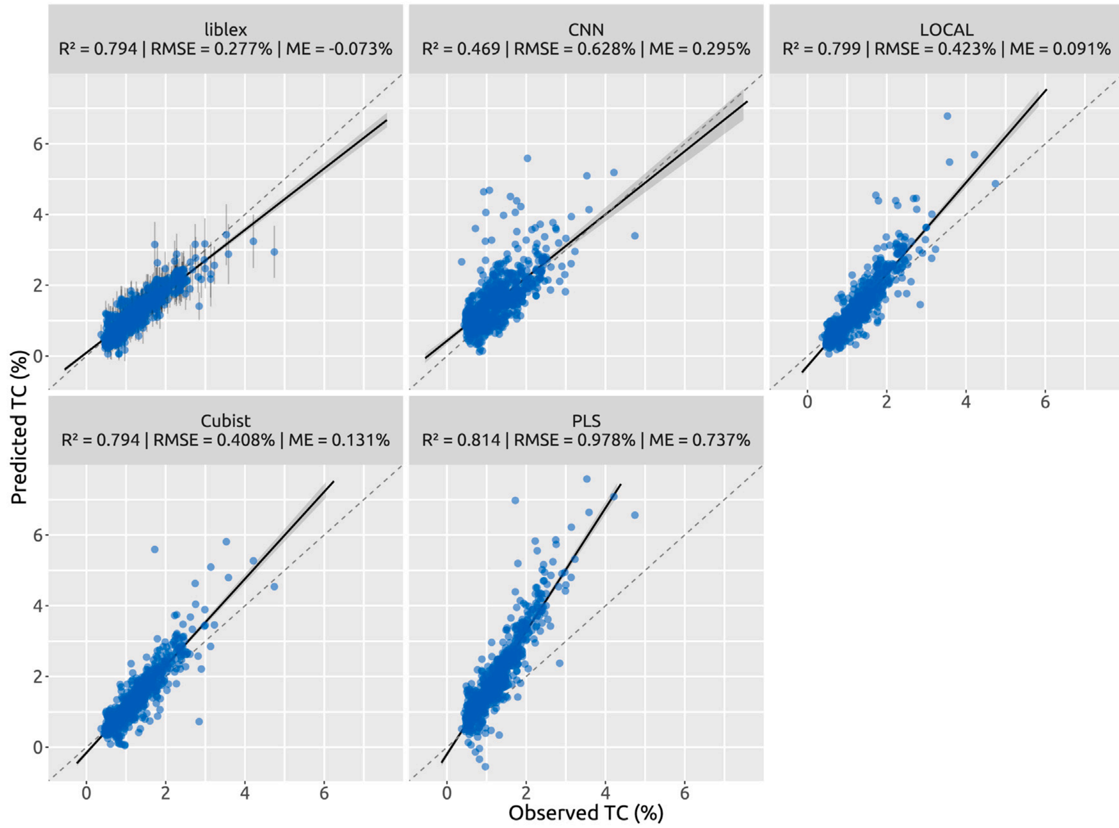

Fig. B.8. Validation plots for `liblex` and benchmarks without target-domain adaptation. The vertical bands in the `liblex` scatter plot indicate the standard deviation across experts for each prediction.

CRediT authorship contribution statement

Leonardo Ramirez-Lopez: Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Maxime Metz:** Writing – review & editing, Conceptualization. **Matthieu Lesnoff:** Writing – review & editing. **Claudio Orellano:** Writing – review & editing, Conceptualization. **Estefania Perez-Fernandez:** Writing – review & editing, Conceptualization. **Marcal Plans:** Writing – review & editing. **Timo Breure:** Writing – review & editing. **Thorsten Behrens:** Writing – review & editing. **Raphael Viscarra Rossel:** Writing – review & editing. **Yi Peng:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors thank Dr Xiaocheng Li for supervising the first author's MSc research report at Imperial College London, Imperial College Business School, from which this work partly originated, and for providing valuable guidance and feedback during the early conceptual development of the study. The authors also thank the Soil Resources Group at ETH Zurich (Switzerland), Institute of Terrestrial Ecosystems (ITES), for funding the data collection in Congo as part of its own research activities and for kindly making the data available for use in this study. The USDA NRCS National Soil Survey Center and the Kellogg Soil Survey Laboratory (NSSC–KSSL) are gratefully acknowledged for curating and maintaining the North American spectral library employed in this study; the SoilSpec4GG initiative is also acknowledged for facilitating access to this resource. Finally, the first author thanks Galo and Pablo for their patience and for keeping him as an anchor within their affective neighbourhood.

Appendix A. Locally weighted average partial least squares (waPLS) used in `liblex`

The local linear regressions in `liblex` are computed using a locally weighted average partial least squares (waPLS) regression. This method, originally proposed within the LOCAL algorithm [8], combines multiple regression vectors fitted with different numbers of PLS components into a single final vector. For a given anchor a with neighbourhood $\mathcal{N}_k(a)$, let $\kappa_{\min}$ and $\kappa_{\max}$ denote the minimum and maximum retained components. For each $\kappa \in \{\kappa_{\min}, \ldots, \kappa_{\max}\}$, a local PLS model is fitted on $\mathcal{N}_k(a)$ with κ components, yielding coefficients $\boldsymbol{\beta}_{a,\kappa}$. The root-mean-square reconstruction error of the anchor spectrum, $s_{a,\kappa}$, is computed using the projection–reconstruction mapping from the fitted model, and the coefficient norm $g_{a,\kappa} = \|\boldsymbol{\beta}_{a,\kappa}\|_2$ is used as a stability proxy. Both diagnostics are then normalised across κ within each anchor, yielding $\tilde{s}_{a,\kappa}$ and $\tilde{g}_{a,\kappa}$. Unnormalised weights are defined as

$$u_{a,\kappa} = \frac{1}{(\tilde{s}_{a,\kappa} + \varepsilon)^{\alpha_s}(\tilde{g}_{a,\kappa} + \varepsilon)^{\alpha_g}},$$

with small ε for numerical stability and exponents α_s, α_g (default = 1). Normalisation gives

$$w_{a,\kappa} = \frac{u_{a,\kappa}}{\sum_{\nu=\kappa_{\min}}^{\kappa_{\max}} u_{a,\nu}},$$

and the final waPLS regression vector for anchor a is

$$\boldsymbol{\beta}_a^{\text{waPLS}} = \sum_{\kappa=\kappa_{\min}}^{\kappa_{\max}} w_{a,\kappa}\, \boldsymbol{\beta}_{a,\kappa}.$$

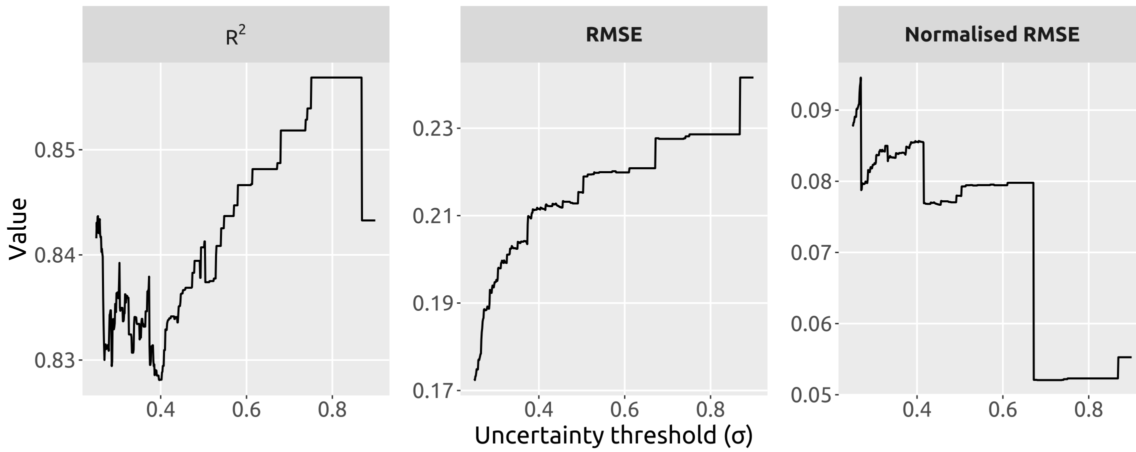

Fig. C.9. Change in prediction performance metrics resulting from the removal of predictions exceeding increasing uncertainty thresholds, where uncertainty is quantified as the standard deviation of predictions across experts.

This formulation penalises models that reconstruct the anchor poorly or produce unstable coefficients, while averaging across component counts to reduce sensitivity to the specific choice of $\kappa_{\min}$ and $\kappa_{\max}$.

Appendix B. Validation plots for the scenario without target-domain adaptation

See Fig. B.8.

Appendix C. Performance metrics vs. uncertainty threshold

See Fig. C.9.

Data availability

The open-source implementation of the `liblex` algorithm is publicly available, together with detailed documentation, through the `resemble` package for the R programming environment. The stable release is available from CRAN at https://CRAN.R-project.org/package=resemble, and the development version is available at https://github.com/l-ramirez-lopez/resemble. Additional documentation and usage examples are provided at https://l-ramirez-lopez.github.io/resemble/.

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